

MATH 2123 - Mathematics III

2019 Regular Examination

Beginner Guide + Topic Basics + Problem-Solving Procedures + Full Solutions

Designed for a beginner

You do not need to be an expert before using this guide. For every question set, first learn the minimum idea, then follow a repeatable procedure, and only then study the complete solution.

Source note

This guide follows the three photographed pages of the 2019 KUET Math 2123 paper. Two scan ambiguities are marked explicitly: Q1(c) has a faint z after (a^2-3) , and Q7(a) omits the vector field in print; the handwritten note is used for Q7(a).

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Quick Topic Map

Set	Main topics	Core skill
1	Linear systems, polynomial interpolation, orthogonal matrix	Translate conditions into equations; rank/consistency
2	Rank, inverse, eigenvalues, diagonalization	Row reduction and characteristic equation
3	Symmetric matrices, Laplace transform, unit-step functions	Match transpose entries; shifting rules
4	Inverse Laplace, improper integral, graph-to-Laplace	Partial fractions; convert graph to steps
5	Triple product, directional derivative, conservative field	Dot/cross/gradient and potential function
6	Vector identity, Green theorem, Stokes/surface integral	Choose the right vector-calculus theorem
7	Volume integral of curl, angle of surfaces, vector operators	Normals, curl, triple bounds
8	Beam deflection, ODE by Laplace transform	Step loads and transform-domain algebra

Minimal Formula Toolbox

- Orthogonal matrix: $A^T A = I$ (equivalently $A^{-1} = A^T$).
- Rank: number of pivots after row reduction; $\det(A) \neq 0$ for a square full-rank matrix.
- Eigenvalues: $\det(A - \lambda I) = 0$; eigenvectors: $(A - \lambda I)v = 0$.
- Laplace: $L\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$.
- Shift: $L\{e^{-at}f(t)\} = F(s+a)$; $L\{u(t-a)g(t-a)\} = e^{-as}G(s)$.
- Directional derivative: $D_u f = \text{grad}(f) \cdot u$, where u is a unit vector.
- Conservative field: $\text{curl}(F) = 0$ in a suitable simply connected region; then $F = \text{grad}(\phi)$.
- Green: $\int_C P dx + Q dy = \iint_D (Q_x - P_y) dA$ for counterclockwise C .
- Stokes: $\iint_S \text{curl}(F) \cdot n dS = \int_{\text{boundary}} F \cdot dr$.
- Angle between surfaces = angle between their normal (gradient) vectors.
- Euler-Bernoulli beam: $EI y^{(4)} = w(x)$, with sign depending on your course convention.

Question Set 1 - Linear systems, polynomial conditions, orthogonal matrices

Topic basics

A homogeneous linear system has all right-hand sides equal to zero. Polynomial conditions such as $P(1)=1$ or $P'(1)=5$ become ordinary linear equations in the unknown coefficients. An orthogonal matrix has mutually perpendicular unit columns, so $A^T A = I$.

Reusable problem-solving procedure

1. Translate every condition into an equation.
2. Solve the resulting linear system carefully.
3. For orthogonality, compute column dot products or $A^T A$.
4. For a parameterized system, compare coefficient-matrix rank with augmented-matrix rank.

1(a) Cubic polynomial from four conditions

Let $P(x)=a+bx+cx^2+dx^3$. Then $P'(x)=b+2cx+3dx^2$.

$$P(1): a + b + c + d = 1$$

$$P'(1): b + 2c + 3d = 5$$

$$P(-1): a - b + c - d = 3$$

$$P'(-1): b - 2c + 3d = 1$$

Add/subtract the equations. From the derivative equations, $4c=4$, so $c=1$. Then $b+3d=3$. From $P(1)$ and $P(-1)$, $a+c=2$ and $b+d=-1$. Thus $a=1$. Solving $b+d=-1$ and $b+3d=3$ gives $d=2$ and $b=-3$.

FINAL ANSWER: $P(x) = 1 - 3x + x^2 + 2x^3$.

1(b) Test whether A is orthogonal

$$A = \left[\begin{bmatrix} 2/3 & 2/3 & 1/3 \end{bmatrix}, \begin{bmatrix} 2/3 & -1/3 & 2/3 \end{bmatrix}, \begin{bmatrix} -1/3 & 2/3 & 2/3 \end{bmatrix} \right]$$

A quick test is to take dot products of columns. The first and third columns are

$$c_1 = (2/3, 2/3, -1/3), \quad c_3 = (1/3, 2/3, 2/3)$$

$$c_1 \cdot c_3 = 2/9 + 4/9 - 2/9 = 4/9 \neq 0$$

Therefore the columns are not mutually perpendicular, so $A^T A$ is not the identity.

FINAL ANSWER: A is NOT an orthogonal matrix.

1(c) Parameter values for no / one / infinitely many solutions

Scan clarification

The mathematically meaningful reading is $x+2y+(a^2-3)z=a$; the z is extremely faint in the photo. Without that z , the coefficient matrix is constant and the system has a unique solution for every a , which would not match the wording of the question.

$$x+2y+z=2; \quad 2x-2y+3z=1; \quad x+2y+(a^2-3)z=a$$

Coefficient determinant:

$$\det A = -6(a-2)(a+2)$$

So for $a \neq 2$ and $a \neq -2$, $\det A \neq 0$ and the solution is unique. At the exceptional values, compare ranks.

- $a=2$: $\text{rank}(A)=\text{rank}([A|b])=2<3$, so infinitely many solutions. One parameter form is $x=1-(4/3)t$, $y=1/2+t/6$, $z=t$.
- $a=-2$: $\text{rank}(A)=2$ but $\text{rank}([A|b])=3$, so the system is inconsistent and has no solution.

FINAL ANSWER: Unique solution for $a \neq \pm 2$; infinitely many for $a=2$; no solution for $a=-2$.

Question Set 2 - Rank, inverse, eigenvalues and diagonalization

Topic basics

Row operations reveal pivots and rank. To find an inverse by row operations, reduce $[A \mid I]$ to $[I \mid A^{-1}]$. Eigenvalues come from $\det(A - \lambda I) = 0$. If we place independent eigenvectors as columns of P , then $P^{-1} A P$ is diagonal.

Reusable problem-solving procedure

1. Compute a determinant first when a square matrix contains a parameter.
2. At determinant-zero values, row-reduce to find the actual rank.
3. For an inverse, work on $[A \mid I]$ and apply every row operation to both sides.
4. For diagonalization, find eigenvalues, then an eigenvector for each, then build P .

2(a) Rank of B depending on k

$$B = \begin{bmatrix} k & 1 & 2 \\ 1 & 1 & 1 \\ -1 & 1 & 1-k \end{bmatrix}$$

$$\det(B) = -(k-2)(k+1)$$

Therefore $\text{rank}(B)=3$ whenever k is not 2 or -1. At $k=2$ and $k=-1$ the determinant is zero. Row reduction still leaves two pivots in both cases, so $\text{rank}=2$.

FINAL ANSWER: $\text{rank}(B)=3$ for $k \neq 2, -1$; $\text{rank}(B)=2$ for $k=2$ or $k=-1$.

2(b) Inverse by elementary row operations

$$A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

Start with $[A \mid I]$. Use $R_2 \leftarrow R_2 - R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$[[1, 3, 3 \mid 1, 0, 0], [0, 1, 0 \mid -1, 1, 0], [0, 0, 1 \mid -1, 0, 1]]$$

Now eliminate the 3s in row 1: $R_1 \leftarrow R_1 - 3R_2 - 3R_3$.

$$[[1, 0, 0 \mid 7, -3, -3], [0, 1, 0 \mid -1, 1, 0], [0, 0, 1 \mid -1, 0, 1]]$$

FINAL ANSWER: $A^{-1} = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$.

2(c) Eigenvalues, eigenvectors and diagonal matrix

$$M = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

$$\det(M - \lambda I) = (1 - \lambda)(3 - \lambda) - 8 = \lambda^2 - 4\lambda - 5$$

$$(\lambda - 5)(\lambda + 1) = 0 \Rightarrow \lambda = 5, -1$$

- For $\lambda = 5$: $(M - 5I)v = 0$ gives $x = y$, so $v_1 = (1, 1)^T$.

- For $\lambda = -1$: $(M + I)v = 0$ gives $x = -2y$, so $v = (-2, 1)^T$.

Choose P with these eigenvectors as columns, for example

$$P = \begin{bmatrix} -2 & 1 \\ 1 & 1 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} -1/3 & 1/3 \\ 1/3 & 2/3 \end{bmatrix}$$

$$P^{-1} M P = \text{diag}(-1, 5)$$

FINAL ANSWER: Eigenpairs: $(-1, (-2, 1)^T)$ and $(5, (1, 1)^T)$; one diagonalizing P is $\begin{bmatrix} -2 & 1 \\ 1 & 1 \end{bmatrix}$.

Question Set 3 - Symmetric matrices, Laplace transform, unit-step functions

Topic basics

A matrix is symmetric when $A=A^T$, so entries reflected across the main diagonal must match. For Laplace transforms, use linearity, the frequency-shift rule, and differentiation in s . Unit-step functions are the clean way to encode changes at specific times.

Reusable problem-solving procedure

1. For symmetry, equate $A_{12}=A_{21}$, $A_{13}=A_{31}$, $A_{23}=A_{32}$.
2. For $t^n f(t)$, use $L\{t^n f\}=(-1)^n d^n F/ds^n$.
3. For $e^{(-at)}f(t)$, replace s by $s+a$.
4. For piecewise functions, begin with the first formula and add the change at every breakpoint.

3(a) Find a, b, c so that A is symmetric

$$A = \begin{bmatrix} 2 & a-2b+2c & 2a+b+c \\ 3 & 5 & a+c \\ 0 & -2 & 7 \end{bmatrix}$$

Symmetry gives three equations:

$$a-2b+2c=3; \quad 2a+b+c=0; \quad a+c=-2$$

Solving them gives $a=11$, $b=-9$, $c=-13$.

FINAL ANSWER: $a=11$, $b=-9$, $c=-13$.

3(b) Laplace transform of $t^2 e^{(-t)} \cos t + (t-1)^2$

First, $L\{\cos t\}=s/(s^2+1)$. Differentiating twice:

$$L\{t^2 \cos t\}=2s(s^2-3)/(s^2+1)^3$$

Multiplication by $e^{(-t)}$ shifts $s \rightarrow s+1$:

$$L\{t^2 e^{(-t)} \cos t\}=2(s+1)((s+1)^2-3)/(((s+1)^2+1)^3)$$

Also, $(t-1)^2=t^2-2t+1$, so

$$L\{(t-1)^2\}=2/s^3 - 2/s^2 + 1/s$$

FINAL ANSWER: $F(s)=2(s+1)((s+1)^2-3)/(((s+1)^2+1)^3) + 1/s - 2/s^2 + 2/s^3$.

3(c) Unit-step form and Laplace transform

$$f(t)=2 \ (0 \leq t < 3); \ 2t \ (3 \leq t < 5); \ t^2/7 \ (t \geq 5)$$

Start with 2. At $t=3$ the change is $2t-2 = 2(t-3)+4$. At $t=5$ the correction from $2t$ to $t^2/7$ becomes, in terms of $\tau=t-5$, $\tau^2/7 - 4\tau/7 - 45/7$.

$$f(t)=2+[2(t-3)+4]u(t-3)+[(t-5)^2/7-4(t-5)/7-45/7]u(t-5)$$

$$L\{f\}=2/s + e^{-3s}(2/s^2+4/s) + e^{-5s}(2/(7s^3)-4/(7s^2)-45/(7s))$$

FINAL ANSWER: The unit-step expression and transform are the two formulas above.

Question Set 4 - Inverse Laplace, improper integral, graph-to-Laplace

Topic basics

Partial fractions split a complicated transform into standard pieces. The convolution theorem says $L^{-1}\{F(s)G(s)\} = (f * g)(t) = \int_0^t f(u)g(t-u)du$. A graph with jumps can be converted directly into unit-step changes.

Reusable problem-solving procedure

1. Factor or complete the square in the denominator.
2. Use partial fractions and rewrite quadratics as $(s+a)^2+b^2$.
3. For improper-integral proofs, introduce a parameter and connect the required integral to a known Laplace integral.
4. For a step graph, list the value changes at each jump and write one step term per change.

4(a) Inverse Laplace by partial fractions

$$F(s) = (5s+3)/[(s-1)(s^2+2s+5)]$$

$$F(s) = 1/(s-1) - (s-2)/(s^2+2s+5)$$

Since $s^2+2s+5 = (s+1)^2+4$ and $-(s-2) = -(s+1)+3$,

$$F = 1/(s-1) - (s+1)/[(s+1)^2+4] + 3/[(s+1)^2+4]$$

FINAL ANSWER: $f(t) = e^t - e^{-t}\cos(2t) + (3/2)e^{-t}\sin(2t)$.

4(b) Prove $\int_0^\infty \sin^2(t)/t^2 dt = \pi/2$

A useful Laplace identity is obtained from $L\{\sin(bt)\} = b/(s^2+b^2)$. Integrating with respect to b from 0 to B gives

$$\int_0^\infty e^{-st} \sin(Bt)/t dt = \arctan(B/s)$$

Let $s \rightarrow 0+$ and $B > 0$. Then

$$\int_0^\infty \sin(Bt)/t dt = \pi/2$$

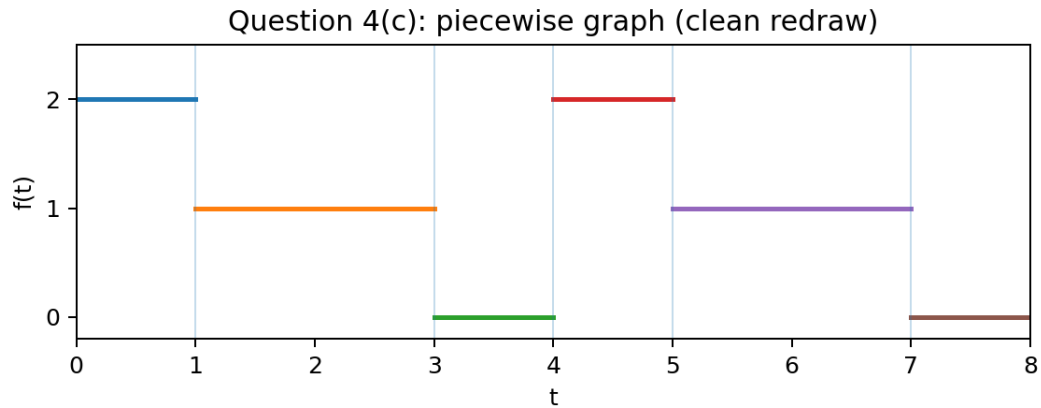
Now define $I(a) = \int_0^\infty \sin^2(at)/t^2 dt$. Differentiate with respect to a :

$$I'(a) = \int_0^\infty \sin(2at)/t dt = \pi/2 \quad (a > 0)$$

Since $I(0) = 0$, we get $I(a) = \pi a/2$. Put $a = 1$.

FINAL ANSWER: $\int_0^\infty \sin^2(t)/t^2 dt = \pi/2$.

4(c) Laplace transform from the graph



Read the graph as: 2 on $[0, 1)$, 1 on $[1, 3)$, 0 on $[3, 4)$, 2 on $[4, 5)$, 1 on $[5, 7)$, and 0 afterward.

The jump changes are -1 at 1, -1 at 3, +2 at 4, -1 at 5, -1 at 7.

$$f(t) = 2 - u(t-1) - u(t-3) + 2u(t-4) - u(t-5) - u(t-7)$$

FINAL ANSWER: $F(s) = [2 - e^{-s} - e^{-3s} + 2e^{-4s} - e^{-5s} - e^{-7s}] / s$.

Question Set 5 - Triple product, directional derivative, conservative field

Topic basics

The scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ gives signed volume; absolute value gives physical volume. A directional derivative is $\text{grad}(f) \cdot \mathbf{u}$ where $\mathbf{u}$ is a unit direction. A conservative field is a gradient field, so mixed partial derivatives agree and a scalar potential can be reconstructed by integration.

Reusable problem-solving procedure

1. For volume, place the three vectors in a determinant and take absolute value.
2. For a directional derivative, normalize the direction first.
3. For conservativity, compare P_y with Q_x , P_z with R_x , and Q_z with R_y .
4. To find a potential, integrate one component and use the others to determine the missing functions.

5(a) Volume of a parallelepiped

$$\mathbf{a}=(2,-3,4), \mathbf{b}=(1,2,-1), \mathbf{c}=(3,-1,2)$$

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \det[[2,-3,4],[1,2,-1],[3,-1,2]] = -7$$

FINAL ANSWER: Volume = $|-7| = 7$ cubic units.

5(b) Directional derivative

$$f=yz+zx+xy, \text{ point}=(1,-1,2), \text{ direction } \mathbf{A}=(3,6,-2)$$

$$\text{grad } f=(z+y, z+x, y+x) \rightarrow \text{at point} = (1,3,0)$$

$$|\mathbf{A}|=7, \text{ so } \mathbf{u}=(3/7, 6/7, -2/7)$$

$$D_{\mathbf{u}} f=(1,3,0) \cdot (3/7, 6/7, -2/7)=21/7=3$$

FINAL ANSWER: Directional derivative = 3.

5(c) Conservative field and potential

$$\mathbf{F}=(e^x \cos y + yz)\mathbf{i} + (xz - e^x \sin y)\mathbf{j} + (xy+z)\mathbf{k}$$

Check the cross partials: $P_y=-e^x \sin y+z=Q_x$, $P_z=y=R_x$, and $Q_z=x=R_y$. Hence $\text{curl}(\mathbf{F})=0$ and $\mathbf{F}$ is conservative.

Integrate P with respect to x :

$$\phi=e^x \cos y + xyz + g(y,z)$$

Matching ϕ_y with Q gives $g_y=0$, so $g=g(z)$. Matching ϕ_z with R gives $g'(z)=z$, hence $g=z^2/2+C$.

FINAL ANSWER: Potential: $\phi=e^x \cos y + xyz + z^2/2 + C$.

Question Set 6 - Vector identity, Green theorem and surface integral of curl

Topic basics

Product rules work for vector differential operators just as they do in ordinary calculus. Green converts a closed line integral in the plane into a double integral. A surface integral of curl can be computed directly or by Stokes; direct projection is convenient for a plane.

Reusable problem-solving procedure

1. Expand vector identities component-by-component once; after that, use the compact rule.
2. For Green, identify P and Q and confirm the direction is counterclockwise.
3. For a plane $z=g(x,y)$, use $n \, dS=(-g_x, -g_y, 1)dA$ for upward orientation.
4. Project the surface onto the xy -plane and integrate over the projected region.

6(a) Verify $\text{div}(\phi A) = \text{grad}(\phi) \cdot A + \phi \text{div}(A)$

$$\phi = e^{xyz}, \quad A = (ax, -by, cz)$$

$$\text{grad}(\phi) = e^{xyz}(yz, xz, xy), \text{ and } \text{div}(A) = a - b + c.$$

$$\text{grad}(\phi) \cdot A = (a - b + c)xyz e^{xyz}$$

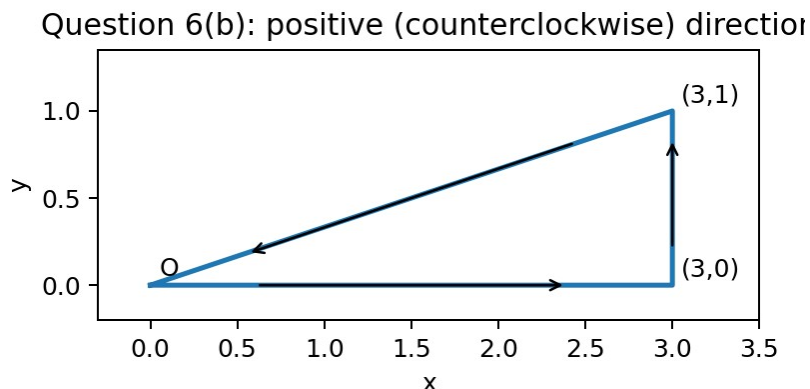
$$\phi \text{div}(A) = (a - b + c)e^{xyz}$$

$$\text{RHS} = (a - b + c)(xyz + 1)e^{xyz}$$

Direct differentiation of $\text{div}(\phi A)$ gives exactly the same expression, so the identity is verified.

FINAL ANSWER: Both sides equal $(a - b + c)(xyz + 1)e^{xyz}$.

6(b) Closed line integral around the triangle



$$A = (y^2 + 2x)\mathbf{i} + (3y - 4x)\mathbf{j}; \quad P = y^2 + 2x, \quad Q = 3y - 4x$$

The indicated direction is counterclockwise, so Green theorem applies with positive sign.

$$Q_x - P_y = -4 - 2y$$

The triangle is $0 \leq x \leq 3$ and $0 \leq y \leq x/3$. Thus

$$\oint_C A \cdot d\mathbf{r} = \int_0^3 \int_0^{x/3} (-4 - 2y) \, dy \, dx = -7$$

FINAL ANSWER: Closed line integral = -7.

6(c) Surface integral of curl over $2x + y + 2z = 6$

$$\mathbf{F} = (x + 2y)\mathbf{i} - 3z\mathbf{j} + x\mathbf{k}$$

$$\text{curl}(\mathbf{F}) = (3, -1, -2)$$

Write $z = 3 - x - y/2$. For upward orientation, $\mathbf{n} \, dS = (1, 1/2, 1) \, dA$. Therefore

$$\text{curl}(\mathbf{F}) \cdot \mathbf{n} \, dS = (3, -1, -2) \cdot (1, 1/2, 1) \, dA = (1/2) \, dA$$

The projection is the triangle $2x + y \leq 6$ in the first quadrant, with intercepts 3 and 6, area 9.

FINAL ANSWER: Surface integral = $9/2$ for the upward normal. Reverse orientation gives $-9/2$.

Question Set 7 - Volume integral of curl, angle between surfaces, vector operators

Topic basics

A triple integral over a tetrahedron needs correct nested bounds. The angle between two surfaces at an intersection point is the acute angle between their normals. Operator expressions such as $(\mathbf{A} \times \text{grad})\phi$ mean apply the derivative operator to ϕ , then form the cross product.

Reusable problem-solving procedure

1. Write the region as inequalities and choose simple nested bounds.
2. Compute curl first; then integrate each component separately.
3. For surface angle, define $F_1=0$ and $F_2=0$, compute both gradients, then use the dot-product angle formula.
4. For $\mathbf{A} \times \text{grad}(\phi)$, compute $\text{grad}(\phi)$ first, then use the determinant formula for a cross product.

7(a) Triple integral of curl over a tetrahedron

Scan clarification

The printed photo gives the region but does not visibly print the vector field. The handwritten annotation beside the question reads approximately $\mathbf{F} = xyz \mathbf{i} + yz^2 \mathbf{j} + y^2 \mathbf{k}$. The following solution uses that field.

$$\mathbf{F} = (xyz, yz^2, y^2)$$

$$\text{curl}(\mathbf{F}) = (2y - 2yz, xy, -xz)$$

Region: $x \geq 0, y \geq 0, z \geq 0, 2x + 2y + z \leq 4$. Convenient bounds are

$$0 \leq x \leq 2, 0 \leq y \leq 2 - x, 0 \leq z \leq 4 - 2x - 2y$$

Integrating the three components over these bounds gives

$$\int \int \int \text{curl}(\mathbf{F}) \, dV = (8/15, 8/15, -16/15)$$

FINAL ANSWER: $= (8/15)\mathbf{i} + (8/15)\mathbf{j} - (16/15)\mathbf{k}$, under the handwritten-field reading above.

7(b) Angle between the two surfaces

$$S_1: x^2 + y^2 + z^2 = 9; \quad S_2: z = x^2 + y^2 - 3; \quad \text{point} = (2, -1, 2)$$

Normals can be taken as gradients:

$$\mathbf{n}_1 = (2x, 2y, 2z) = (4, -2, 4)$$

$$\mathbf{n}_2 = \text{grad}(z - x^2 - y^2 + 3) = (-2x, -2y, 1) = (-4, 2, 1)$$

$$\cos(\theta) = |\mathbf{n}_1 \cdot \mathbf{n}_2| / (|\mathbf{n}_1| |\mathbf{n}_2|) = 16 / (6 \sqrt{21}) = 8 / (3 \sqrt{21})$$

FINAL ANSWER: $\theta = \cos^{-1}\left[\frac{8}{3\sqrt{21}}\right]$ (acute/actual angle).

7(c) Compute $(\mathbf{A} \times \text{grad})\phi$ and $\mathbf{A} \times (\text{grad } \phi)$

$$\mathbf{A} = (2yz, -x^2y, xz^2), \quad \phi = 2x^2yz^3$$

$$\text{grad}(\phi) = (4xyz^3, 2x^2z^3, 6x^2yz^2)$$

Because the differential operator acts on the scalar ϕ to its right, $(\mathbf{A} \times \text{grad})\phi = \mathbf{A} \times (\text{grad } \phi)$. Their common value is

$$[-2x^3z^2(3xy^2+z^3)]\mathbf{i} - [4x^2yz^3(3y-z^2)]\mathbf{j} + [4x^2yz^3(xy+z)]\mathbf{k}$$

FINAL ANSWER: Both requested expressions are equal to the vector shown above.

Question Set 8 - Beam deflection and differential equation by Laplace transform

Topic basics

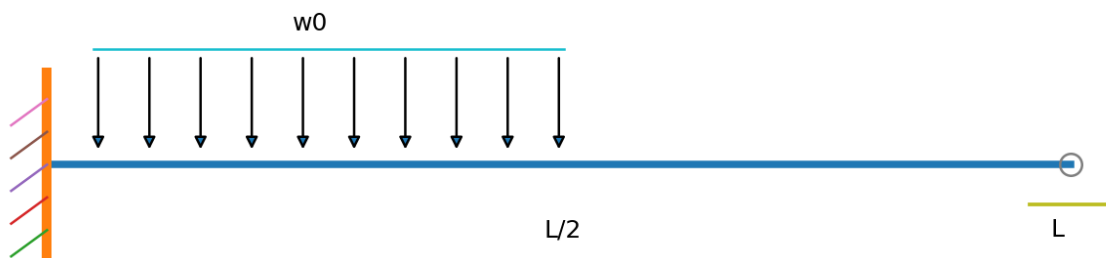
For Euler-Bernoulli beams, load is related to the fourth derivative of deflection. A discontinuous load is easiest to write with a unit-step (Macaulay) term. For an ODE, Laplace transforms convert derivatives and initial conditions into an algebraic equation for $Y(s)$.

Reusable problem-solving procedure

1. Write the load using a step function.
2. Integrate $EI y^{(4)} = w(x)$ four times, introducing constants.
3. Apply all four beam boundary conditions.
4. For an ODE, transform every derivative, insert initial values immediately, solve for $Y(s)$, then invert.

8(a) Deflection of a fixed-pinned beam with half-span UDL

Question 8(a): fixed-pinned beam, load on first half



Take downward deflection/load as positive. The load is

$$w(x) = w_0[1 - u(x - L/2)]$$

Using $EI y^{(4)} = w(x)$ and Macaulay brackets $\langle x - a \rangle^n$ (zero for $x < a$, $(x - a)^n$ for $x \geq a$), four integrations give

$$EI y = w_0 x^4/24 - w_0 \langle x - L/2 \rangle^4/24 + C_1 x^3/6 + C_2 x^2/2 + C_3 x + C_4$$

Boundary conditions: fixed end at $x=0$ gives $y(0)=0$ and $y'(0)=0$, hence $C_4=C_3=0$. The simple support at $x=L$ gives $y(L)=0$ and bending moment $M(L)=EI y''(L)=0$.

Solving the last two conditions:

$$C_1 = -(57/128)L w_0, \quad C_2 = (9/128)L^2 w_0$$

$$y(x) = w_0/EI [x^4/24 - \langle x - L/2 \rangle^4/24 - 19Lx^3/256 + 9L^2x^2/256]$$

Equivalent piecewise form:

$$0 \leq x \leq L/2: y = w_0/EI [x^4/24 - 19Lx^3/256 + 9L^2x^2/256]$$

$$L/2 \leq x \leq L: y = w_0/EI [-L^4/384 + L^3x/48 - 7L^2x^2/256 + 7Lx^3/768]$$

FINAL ANSWER: Deflection is given by the Macaulay or piecewise formula above. If your course uses $EI y^{(4)} = -w$, reverse the overall sign.

8(b) Solve $y'' + 2y' + 5y = e^{-t} \sin t$, $y(0) = 0$, $y'(0) = 1$

Let $Y(s) = \mathcal{L}\{y\}$. Transforming:

$$(s^2 Y - 1) + 2(sY) + 5Y = 1/[(s+1)^2 + 1]$$

$$Y = [1 + 1/((s+1)^2 + 1)] / [(s+1)^2 + 4]$$

Put $u = s+1$ and decompose:

$$Y = 1/(u^2 + 4) + 1/[(u^2 + 1)(u^2 + 4)]$$

$$1/[(u^2 + 1)(u^2 + 4)] = (1/3)[1/(u^2 + 1) - 1/(u^2 + 4)]$$

Therefore

$$Y = (1/3)/(u^2 + 1) + (2/3)/(u^2 + 4)$$

FINAL ANSWER: $y(t) = e^{-t}[\sin t + \sin(2t)]/3$.

Rapid Revision - What to do when you see each problem type

Problem type	Fast procedure
Polynomial conditions	Write one equation for each $P(a)$ or $P'(a)$; solve coefficients.
Orthogonal matrix	Check $A^T A = I$ or column dot products.
Parameterized linear system	$\det \neq 0 \rightarrow$ unique; $\det = 0 \rightarrow$ compare $\text{rank}(A)$ and $\text{rank}([A b])$.
Rank with parameter	Factor determinant; row-reduce only at exceptional parameter values.
Inverse	Augment with I and row-reduce to I .
Eigenvalues	$\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$.
Symmetric matrix	Mirror entries across main diagonal.
Laplace with t^n	Differentiate $F(s)$ n times and apply sign $(-1)^n$.
Step functions	Initial formula + change at each breakpoint.
Inverse Laplace	Partial fraction + complete square + shifting.
Directional derivative	$\text{grad } f$ at point, normalize direction, dot.
Conservative field	Check $\text{curl} = 0$; integrate one component to get potential.
Green theorem	$Q_x - P_y$ over planar region; watch orientation.
Surface curl integral	Compute curl, choose $n \, dS$, project surface.
Angle of surfaces	Angle between gradient normals.
Beam deflection	$EI y^{(4)} = w$; integrate 4 times; apply 4 boundary conditions.
ODE by Laplace	Transform, apply initial data, solve $Y(s)$, invert.

Final Answers at a Glance

- 1(a) $P = 1 - 3x + x^2 + 2x^3$. 1(b) A is not orthogonal. 1(c) unique for $a \neq \pm 2$; infinite at $a = 2$; none at $a = -2$.
- 2(a) rank 3 except $k = 2, -1$, where rank 2. 2(b) $A^{-1} = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$. 2(c) $\lambda = -1, 5$.
- 3(a) $a = 11, b = -9, c = -13$. 3(b) transform as derived. 3(c) step transform as derived.
- 4(a) $e^t - e^{-t} \cos 2t + (3/2)e^{-t} \sin 2t$. 4(b) $\pi/2$. 4(c) $[2e^{-s} - se^{-3s} + 2e^{-4s} - e^{-5s} - 7e^{-7s}]/s$.
- 5(a) 7. 5(b) 3. 5(c) $\phi = e^x \cos y + xyz + z^2/2 + C$.
- 6(a) both sides $= (a - b + c)(xyz + 1)e^{xyz}$. 6(b) -7. 6(c) $9/2$ upward.
- 7(a) $(8/15, 8/15, -16/15)$ under the handwritten F reading. 7(b) $\cos^{-1}[8/(3\sqrt{21})]$. 7(c) common cross-product vector as shown.
- 8(a) $y = w_0/EI[x^4/24 - xL/2 + L^4/24 - 19Lx^3/256 + 9L^2x^2/256]$. 8(b) $y = e^{-t}(\sin t + \sin 2t)/3$.

Exam strategy

Do not try to memorize entire solutions. Memorize the procedure for each problem type. In the exam, first write the governing formula/theorem, then substitute the given data, show the essential algebra, and box the final answer.